Friction Stir Welding in Wrought and Cast Aluminum Alloys: Weld Quality Evaluation and Effects of Processing Parameters on Microstructure and Mechanical Properties

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Friction stir welding (FSW) is a solid-state process widely used for joining similar and dissimilar materials for critical applications in the transportation sector. Understanding the effects of the process on microstructure and mechanical properties is critical in design for structural integrity. In this study, four aluminum alloy systems (wrought 6061-T651 and cast A356, 319, and A390) were processed in both as-fabricated and pre-weld heat-treated (T6) conditions using various processing parameters. The effects of processing and heat treatment on the resulting microstructures, macro-/micro-hardness, and tensile properties were systematically investigated and mechanistically correlated to changes in grain size, characteristic phases, and strengthening precipitates. Tensile tests were performed at room temperature both along and across the welding zones. A new method able to evaluate weld quality (using a weld quality index) was developed based on the stress concentration calculated under tensile loading. Optimum processing parameter domains that provide both defect-free welds and good mechanical properties were determined for each alloy and associated with the thermal history of the process. These results were further related to characteristic microstructural features, which can be used for component design and materials/process optimization.

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I. INTRODUCTION AND LITERATURE REVIEW

FRICTION stir welding (FSW) is a complete solid-state welding technique, introduced in 1991 by Thomas *et al.*[1,2] at The Welding Institute in the U.K. The process is used for both similar and dissimilar metals and alloys, producing welds with very few defects, such as shrinkage porosity, that is often found in parts joined by fusion welding.[3] Understanding its fundamentals and advantages on weld quality and mechanical properties has been a subject of investigation in numerous studies. Arbegast *et al.*[4] addressed weld quality describing several types of large-scale welding defects, and among those, the "wormhole" and "surface lack of fill" were the two most important defects near the welding line. These specific defects are primarily associated with voids, cavities, and lack of material in local areas, as well as surface roughness. The potential causes of these welding issues are related to overly hot or cold processing conditions, and some other factors related to the workpiece. The weld quality is strongly affected by processing parameters, which are dependent on the type of working materials.[5] Hot or cold processing defects are caused by excessive rotation and/or traverse speed combinations. Kim *et al.*[6] studied the effects of processing parameters on the weld quality of die-cast Al ADC12 (Al-Si-Cu-Mg-Zn) alloy and proposed three types of FSW defects: (1) a large mass of flash due to the excess heat input; (2) cavity or groove-like defect caused by an insufficient heat input; and (3) cavities produced by abnormal stirring. Doude *et al.*[7] also found that the same defects formed in wrought Al AA2219 alloy at high rotation and traverse speeds. Tutunchilar *et al.*[8] and D'Urso *et al.*[9] reported possible defect formation mechanisms using finite element analysis (FEA) simulation models. The material flow in FSW is limited by insufficient heat input, which results in voids and cavities forming closer to the advancing side of the weld (AS, where tool rotates in the same direction to the traverse direction) rather than the retreating side (RS, where tool rotates in opposite direction to the traverse direction). The results have consistently shown in all these studies that the energy input and stirring rate are strongly affecting defects formation, and their proper selection is key to controlling weld quality. However, no quantitative models have been found in the literature relating processing parameters to the weld quality or types and locations of defects.

Microstructure refinement occurs during mechanical stirring in the dynamically recrystallized zone (DXZ or nugget).[2–4] This is an another important effect contributing to improvements in weld quality and application performance. Numerous studies[4,10–14] discussed

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the intense plastic deformation of DXZ during FSW, resulting in much smaller grain sizes, as well as more refined and uniform distribution of eutectic and other secondary phases. Grain refinement is determined by local dynamic recrystallization, which is affected by both strain rates and thermal conditions, and the final size varies in different location within the nugget.[14–18] Singh *et al.*[14] reported in a study on wrought Al 6082 alloy that the recrystallized nugget grain size depends on the combinations of rotation and traverse speeds, and the corresponding thermal energy inputs. Based on their observation, the resulting heat input plays the most important role in controlling the grain structure. In cast Al alloys, both grain and secondary phase refinement occur in the nugget zone.[19,20] Ma *et al.*[19] studied the eutectic Si particles in the nugget zone. Their study indicated that the size of the Si was reduced and more uniformly distributed in the nugget zone. They also found that at different locations within the nugget, different distribution patterns and sizes would occur depending on the local material flow characteristics. Previous studies on Al systems[21] indicate very different FSW responses of the alloys, depending on their specific processing conditions and microstructures. Thus, it is important to study both individual and combined effects of chemical composition, microstructure, and pre-weld heat treatment in order to fully understand the evolution mechanisms in FSW.

It was also reported that subgrain structures, including dislocations and subgrain boundaries introduced by mechanical stirring, are significantly refined during dynamic recrystallization in the nugget zone.[22,23] For precipitation-strengthened alloys, the evolution of precipitates was often investigated. In both DXZ and heat-affected zone (HAZ), the structure and coherency of the original precipitates are affected by FSW.[24–27] It was also found that in DXZ, precipitate phases dissolved into the matrix and re-precipitated with the increased temperature during FSW, while in HAZ, usually precipitate coarsened.[28] Even if the precipitate evolution mechanisms are the same, their final structures will differ for different types of precipitates (*e.g.,* β'' in Al-Mg-Si alloy[26] and θ'' in Al-Cu alloy[27]), due to their different sensitivities to temperature increase. Olea *et al.*[25] investigated precipitates evolution in 2xxx (Al-Cu), 6xxx (Al-Mg-Si), and 7xxx (Al-Zn) before and after FSW. They reported less degradation of the precipitates in FSW HAZ compared to traditional fusion welding, as peak temperature in FSW typically than 80 pct of the melting temperatures of the alloys.

Changes in grain and secondary phase structures influence both the hardness and tensile properties of the materials. As reported by many groups,[19,29–31] improvements in hardness (compared to HAZ) and ductility in the refined DXZ after FSW were observed, primarily due to grain refinement.[10–14] However, hardness and tensile strength within the nugget zone are lower than those of the base material due to the loss of precipitate strengthening as reported in the literature for wrought Al alloys.[25] In cast Al alloys (*e.g.,* A356-F), the increase in hardness and tensile strength in the nugget is additionally correlated to the refinement and redistribution of Si particles and other secondary phases.[19,32] Tensile properties perpendicular to the traverse direction (across the weld) have also been analyzed. In this case, the tensile strength and ductility of wrought Al alloys are in most cases limited by the HAZ due to the coarsening of the precipitates.[10–14,24–28] In cast alloys, increases in tensile strength were observed in some cases, resulting from elimination of porosity and more uniformly distributed secondary phases.[6,19,20] Liu *et al.*[33] found the tensile properties across the welds in Al 2017 to be lower than that of the base material. The revolutionary pitch (the ratio of rotation/traverse speed) was examined and found to determine the weak position and the fracture location during FSW. As the revolutionary pitch exceeds the critical value for a specific material, not only it will generate defects, but also will likely lower tensile properties. The bonding strength between different zones depends on several factors: (1) defects generated within mechanically stirred zones, (2) mixing between different weld zones, and (3) initial defects in base materials. Simar *et al.*[34] developed an integrated model for 6xxx series alloys relating process conditions to mechanical properties. The model consisted of a thermal model of temperature change during FSW, a precipitate growth model, as well as a finite element model coupled with damage mechanisms for tensile property evaluations. The processing-microstructure-property interrelations are predicted with the integrated model. This demonstrates that systematic studies of the microstructural and mechanical property changes for different alloy systems under various processing parameters are important and deserve further consideration and focus.

In this study, experimental investigations were carried out on different wrought and cast Al alloys with and without pre-weld heat treatment, processed with various combinations of rotation and traverse speeds. A new quality index model was developed to qualify the welds based on the morphology of the defects and the stress concentration associated with them. Proper FSW domains were determined for each alloy depending on the quality index classifications. Grain size, band spacing, and secondary phases were characterized and associated with processing parameters. Macro-/micro-hardness and characteristic precipitates were evaluated across the welds, and tensile properties both within and across the nugget were interpreted based on processing conditions and resulting heat input.

II. EXPERIMENTAL METHODOLOGY

A. *Materials and Microstructures*

The Al alloys used in this study for the FSW experiments were wrought 6061-T651 and cast A356, 319, and A390 (A356 and 319 were FSW in both as-cast/F and T6 conditions, and A390 was processed in as-cast/F condition). Chemical compositions of the alloys are listed in Table I.

The selected materials have major chemical differences in three elements: Si, Mg, and Cu. These

Table I. Chemical Compositions of the Studied Al Alloys in Wt Pct

Alloy	Elements				
	Al	Si	Mg	Cu	Cr
6061	base	0.4 to 0.8	0.8 to 1.2	0.15 to 0.4	0.04 to 0.35
A356	base	6.5 to 7.5	0.25 to 0.45	<0.2	
319	base	5.5 to 6.5	<0.1	3 to 4	
A390	base	16 to 18	0.45 to 0.65	4 to 5	

differences will shed light on the effects of Si content, precipitate systems (β'- and β''-Mg_2Si, θ'- and θ''-Al_2Cu, *etc.*), and distribution and morphology of secondary phases. The microstructures were identified in each alloy before FSW, shown in Figures 1 and 2. The size of secondary dendrite arm spacing (SDAS) and eutectic/primary Si particles were also measured as shown in Table II. In all as-cast Al alloys, Figures 1(a) through (c), Si particles are unmodified. After T6 pre-weld heat treatment, the Si particles in A356 and 319, Figures 2(b) and (c), reduced in size due to thermal modification.

B. *Friction Stir Welding Experiments*

All materials were machined into plates with the dimensions $L \times W \times H = 203$ mm $\times$ 127 mm $\times$ 18 mm. The wrought 6061 Al alloy was FSW in the T651 condition. The cast A356 and 319 alloys were FSW in both as-cast (F) and with a pre-weld T6 heat treatment, and A390 was welded in F condition only. The pre-weld heat treatment was used to improve the initial materials' strength and understand precipitate growth during processing. Different T6 heat treatments were used for each alloy, as appropriate, and the conditions are listed in Table III.

The FSW experiments were conducted in room temperature air environment. The welding fixture is shown in Figures 3(a) and (b). The threaded FSW tool is made of AISI H13 steel with a plain shoulder and a conical helix pin. The shoulder has a diameter of 18 mm, and the pin has a length of 13 mm and a diameter of 10 mm, Figures 3(c) and (d). The total penetration depth during FSW was 13 mm.

A single pass of FSW was used for all experiments. Advancing and retreating sides were marked based on the tool rotation and traverse directions. The tool rotation speed ranged from 600 to 1600 RPM for the wrought alloys and from 800 to 1200 RPM for all cast alloys. Tool traverse speeds were varied from 1 to 3 mm/s for all hypoeutectic alloys, and from 0.17 to 0.64 mm/s for the hypereutectic alloy due to its high Si content. The tilt angle of the FSW tool was kept zero in all experiments.

III. RESULTS AND DISCUSSION

A. *Weld Quality Evaluation*

The weld quality index system is based on the stress concentration at the defect tip, when a general tensile loading is applied. In the case of a defect-free weld, the stress will be uniformly distributed in the nugget. However, when defects exist, the defect tip will be under

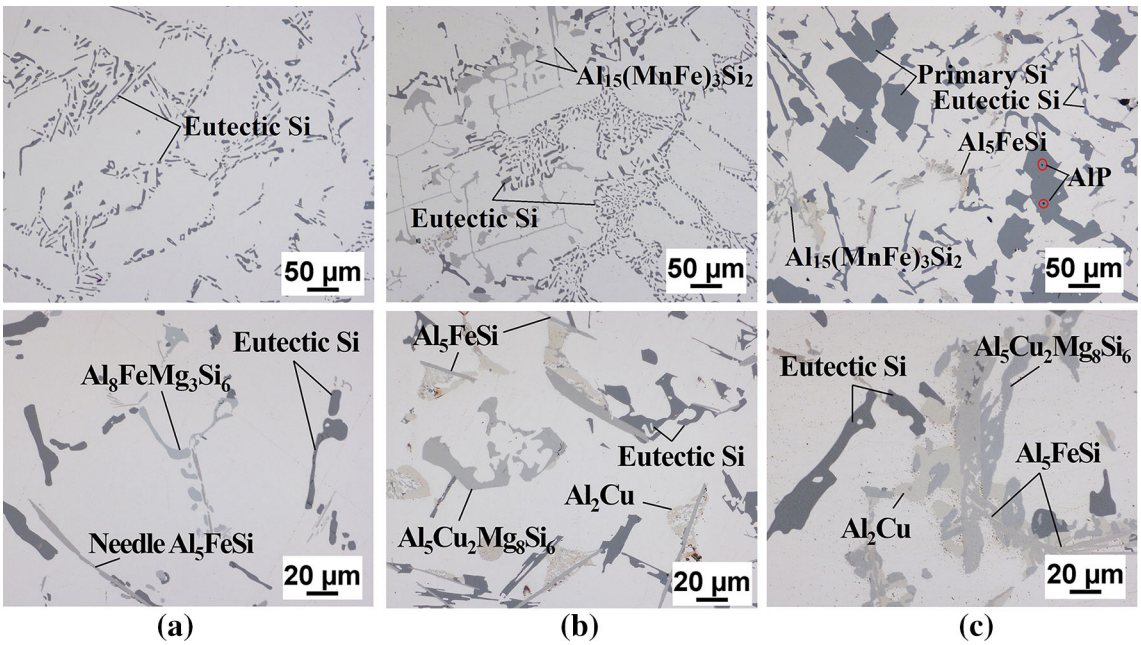


Fig. 1—Microstructures at different magnifications of (*a*) A356-F, (*b*) 319-F, and (*c*) A390-F, showing their characteristic secondary phases.

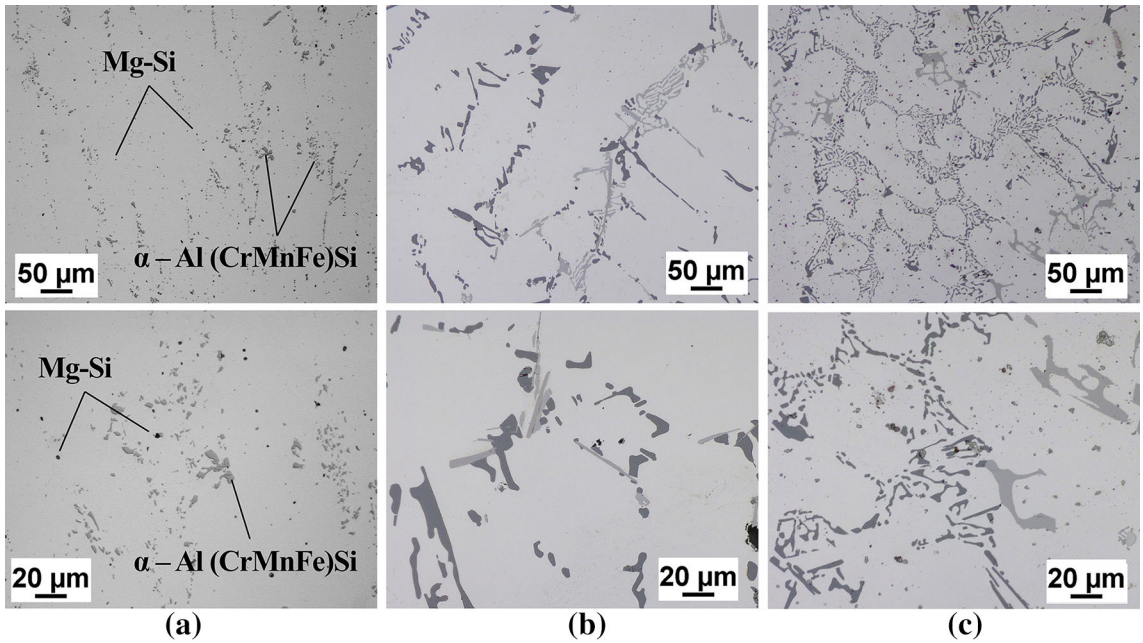


Fig. 2—Microstructures at different magnifications of (*a*) 6061-T651, (*b*) A356-T6, and (*c*) 319-T6 Al alloys, showing their characteristic secondary phases after heat treatment.

Table II. Measurements of Characteristic Microstructural Features of the Base Materials

				Si Particle Size (μm)		
				Eutectic		
	Grain Size (μm)	SDAS (μm)	Porosity (Pct)	Average Size	Aspect Ratio*	Primary
A356-F	694	60	1.05	43	0.4	N/A
319-F	801	63	1.2	55	0.4	N/A
A390-F	751	N/A	1.3	60	0.3	96
6061-T651	400 × 80	N/A	N/A	N/A		N/A
A356-T6	700	64	N/A	36	0.2	N/A
319-T6	810	48	N/A	21	0.3	N/A

*Aspect ratio is calculated using width/length for each particle.

Table III. Pre-Weld T6 Heat Treating Conditions for the Studied Alloys

	Solutionizing T (K/°C)	Solutionizing t (h)	Aging T (K/°C)	Aging t (h)
6061	T651 done at rolling mill facility			
A356	813 K (540 °C)	1.5	426 K (153 °C)	12 (Natural) + 12 (Artificial)
319	768 K (495 °C)	4	453 K (180 °C)	12 (Natural) + 4 (Artificial)

higher stress concentration than the surrounding locations. The amplification of the stress is related to both the size and morphology of the defect. In this case, the amplification of the stress concentration can be used as an indicator of the weld quality. The quality index varies between 0 (poor) to 1 (perfect), and it represents a way to evaluate the safety factor of the welds.

$$\mathrm{QI} = (1 - f) \cdot \bar{M}. \quad [1]$$

The quality index of the weld depends on both the size (defect area fraction, f) and shape (average morphology coefficient, $\bar{M}$) of the defects. The value of f is calculated by the ratio of defect area to weld area as shown in Eq. [2].

$$f = \mathrm{Area}_{\mathrm{Defects}}/\mathrm{Area}_{\mathrm{Weld}}. \quad [2]$$

Average morphology coefficient $\bar{M}$ is related to the orientation of the defect and peak stress concentration when a load is applied.[35] Defects are assumed to be elliptical to simplify the calculations of the stress concentration factors. When defects are subjected to a

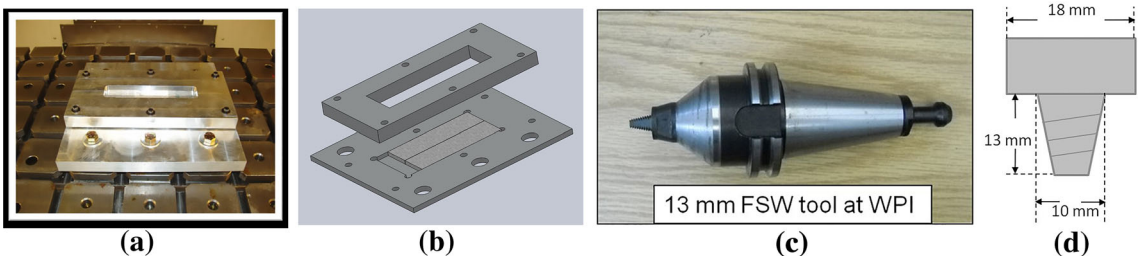


Fig. 3—(*a*, *b*) Fixture used for FSW and (*c*, *d*) FSW tool geometry and dimensions.

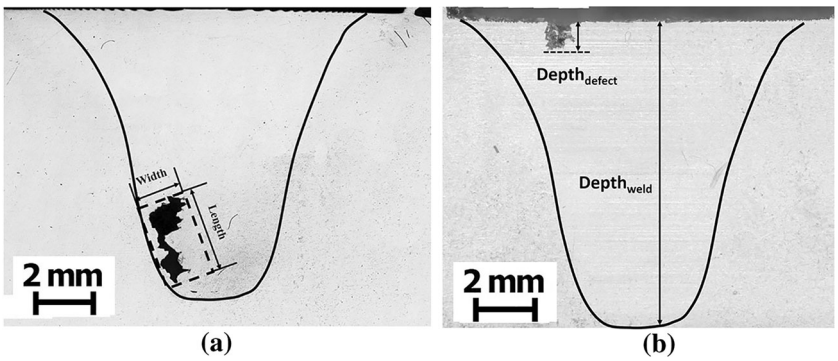


Fig. 4—Characteristic parameters used in the weld quality index model for (*a*) internal tunnel and (*b*) surface defects.

general tension load, the stress at the defect tip is evaluated using Eq. [3].[35]

$$\sigma = \sigma_0 e^{2\alpha_0}\left[\frac{\left(1+e^{-2\alpha_0}\right)\sinh(2\alpha_0)}{\cosh(2\alpha_0)-\cos(2\theta)}-1\right]. \qquad [3]$$

Stresses at the defect tip can be then calculated based on Eq. [3] together with defect measurements obtained from micrographs using image analysis, as indicated in Figure 4. The angle α_0 is defined by the elliptical semi-major axis, a, elliptical semi-minor axis, b, and half focus, c. The parameters are related as $a = c \cdot \cosh(\alpha_0)$ and $b = c \cdot \sinh(\alpha_0)$.

The corresponding stress concentration factor K_{tg} is calculated as the ratio of maximum and minimum stresses at the defect tip, Eq. [4].

$$K_{tg} = 1 + 2\left(\frac{b}{a}\right). \qquad [4]$$

As indicated by Eq. [4], the lowest $K_{tg}$is obtained when $b = a$, and the stress concentration factor increases with the difference between a and b. Thus, a more favorable morphology coefficient will be obtained in the case of a round defect compared to an elongated one.

Irregular defect shapes are usually encountered in actual processing, and in such cases, a defect might contain tips in multiple directions. The morphology coefficient is thus determined using the defect aspect ratio considering its most elongated direction. Subsequently, the morphology coefficient of internal tunnel defects is calculated by the square root of its aspect ratio measured from micrographs, as shown in Figure 4(a).

$$M_T = \sqrt{\frac{\text{Width}}{\text{Length}}}. \qquad [5]$$

In Eq. [5], *Length* is measured in the elongated direction and *Width* perpendicular to it. Thus, the theoretical M_T varies from 0 (needle-like defect) to 1 (circular defect).

Surface defects are considered as half of an internal defect. They have similar but slightly higher stress concentration factors compared to internal defects, which increase when the curvature radius at the tip decreases. Therefore, the surface morphology coefficient is also defined by the aspect ratio, using the relative distances from the surface, Eq. [6].

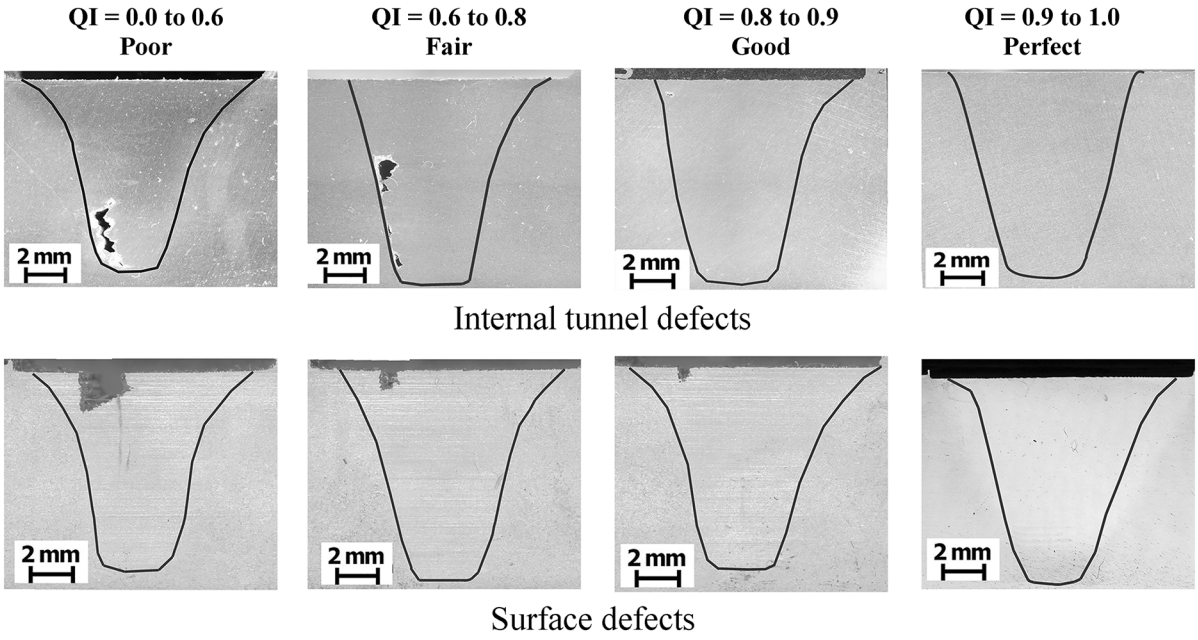
$$M_s = 1 - \left(\frac{\text{Depth}_{\text{Defect}}}{\text{Depth}_{\text{Weld}}}\right)^2. \qquad [6]$$

The average morphology coefficient, $\bar{M}$, for welds containing multiple defects with different geometries and locations is defined to include contributions from both internal tunnel defects (M_T) and surface defects (M_s), as given by Eq. [7].

$$\frac{1}{\bar{M}} = \sum_n \frac{1}{M_{T_n}} \cdot R_n + \sum_m \frac{1}{M_{S_m}} \cdot R_m. \qquad [7]$$

The equation balances the effects of each defect, taking into account its size, shape, and location. In Eq. [7], n and m are the numbers of internal tunnel and

Table IV. Weld Quality Classifications Based on Quality Index Values and Examples of Corresponding Internal Tunnel and Surface Defects Generating Them



surface defects, respectively. R_n and R_m are single defects area fractions as defined in Eq. [8].

$$R_n, R_m = \frac{A_{\text{defect}}}{A_{\text{Total defects}}}. \quad [8]$$

Friction stir welding eliminates pre-existing defects (such as porosity in the cast alloys), and therefore, the nugget zone will only contain defects generated by inadequate stirring during processing. Thus, the quality index calculated using Eq. [1] is suitable for different materials and processing conditions.

B. *Optimization of Processing Parameter Domains*

All welds in this study were characterized according to their quality index and classified into four categories as the weld quality increases: poor, fair, good, and perfect. These weld evaluations are shown and generically exemplified in Table IV.

Processing parameters tested for all studied alloys and conditions are shown in Table V, and "appropriate processing domains" were identified for each alloy. In the as-cast alloys A356-F, 319-F, and A390-F, most of the good and perfect welds occurred at rotation speeds of 800 to 1000 RPM. The narrow appropriate processing parameter domains are due to the low ductility of these alloys. Outside the domain, low rotation speeds may cause internal cavities and high rotation speeds can lead to the severe surface "lack-of-fill" defects. In Figures 5(a) through (c), both internal and surface defects can be observed on the welds produced using processing parameters outside the optimal domain.

For 6061-T651, the proper studied rotation speeds range from 800 to 1600 RPM combined with selected traverse speeds from 1 to 3 mm/s. Due to better ductility, this alloy has a wider appropriate processing parameter domain compared to the cast alloys. The cross section of the welds shown in Figure 5(d) was welded at constant traverse speed, 2 mm/s, using different rotation speeds. No large defects were observed. With more heat input under higher rotation speed, the HAZ became wider, as noticed after polishing.

Weld quality indices of both A356 and 319 are high only for limited combinations of rotation and traverse speeds (*i.e.*, narrow processing domains for good and perfect welds), but they significantly improve after the T6 pre-weld heat treatment, Figures 5(a) and (b) *vs* Figures 5(e) and (f). For A356-T6, the tested processing domains (rotation speeds from 600 to 1200 RPM and traverse speeds from 1 to 3 mm/s) result in good and perfect welds. In 319-T6, weld quality also improves compared to the as-cast FSW conditions, but less than in A356-T6. These observations indicate that pre-weld heat treatment can improve weld quality and enlarge the proper processing domains of the cast alloys.

For alloys 319-F/T6 and A390-F, a triangular distribution of proper processing parameters was observed. This can be explained using the heat transfer model of the FSW process developed by Pan and Lados[36] to relate processing parameters with the heat generation rate and high temperature exposure time in the weld nugget. The model predicts the amount of heat energy that is required to obtain good and perfect welds *via* optimized combinations of tool rotation and traverse speeds. The lower limits of heat input can be calculated

Table V. Processing Domains for Alloys Without and With Pre-Weld Heat Treatment (Optimal Conditions Have Been Italicized)

	Rotation Traverse	600 RPM	800 RPM	1000 RPM	1200 RPM	1400 RPM	1600 RPM
A356 (F)	1 mm/s	0.47-poor	0.53-poor	0.70-fair	*0.88-good*		
	2 mm/s	0.70-fair	*0.84-good*	*0.95-perfect*	0.73-fair		
	3 mm/s	0.61-fair	*0.91-perfect*	0.62-fair	0.54-poor		
319 (F)	1 mm/s	0.51-poor	0.75-fair	0.65-fair	0.58-poor		
	2 mm/s	0.73-fair	*0.96-perfect*	*0.93-perfect*	0.65-fair		
	3 mm/s	0.72-fair	0.62-fair	*0.89-good*	0.67-fair		
A390 (F)	0.17 mm/s		*0.89-good*	*0.94-perfect*	0.41-poor		
	0.25 mm/s		0.47-poor	*0.96-perfect*	0.74-fair		
	0.42 mm/s		0.56-poor	0.67-fair	0.58-poor		
	0.64 mm/s		0.43-poor	0.63-poor	0.51-poor		
6061 (T651)	1 mm/s		*0.93-perfect*	*0.98-perfect*	*0.96-perfect*	*0.95-perfect*	*0.89-good*
	2 mm/s			*0.99-perfect*	*0.93-perfect*	*0.94-perfect*	*0.93-perfect*
	3 mm/s		*0.95-perfect*	*0.96-perfect*	*0.95-perfect*	*0.98-perfect*	*0.96-perfect*
A356 (T6)	1 mm/s	*0.98-perfect*	*0.95-perfect*	*0.96-perfect*			
	2 mm/s	*0.97-perfect*	*0.95-perfect*	*0.95-perfect*	*0.96-perfect*		
	3 mm/s	*0.89-good*	*0.96-perfect*	*0.93-perfect*			
319 (T6)	1 mm/s	*0.85-good*	*0.93-perfect*	0.78-fair			
	2 mm/s	0.73-fair	*0.89-good*	*0.86-good*	0.58-poor		
	3 mm/s	0.70-fair	0.76-fair	*0.83-good*			

using the highest advancing per revolution rate according to Table V.

C. *Microstructures of the Welds*

Grain structures and characteristic phases in the welds of the cast alloys without pre-weld heat treatment (*i.e.*, FSW in as-cast conditions) are shown in Figure 6. For all alloys, the nugget grain sizes were reduced after FSW compared to those of the base materials due to the dynamic recrystallization effects. In all cast alloys, A356-F, 319-F, and A390-F, grain size did not change significantly with different combinations of processing parameters. In this study, all cast alloys were FSW at low rotation speeds (800 to 1000 RPM), thus the total heat input during processing was low, resulting in low recrystallization and grain growth rates. Grains are generally also more refined under higher traverse speeds due to higher deformation rates, which lower the ductility and facilitate the breaking down of the original grains. The presence of the secondary phases, such as eutectic and primary Si particles, also limits the nugget grain growth.

The size of the Si particles (both eutectic and primary) was significantly reduced relative to their original sizes. In A356-F, Figures 6(a) through (c), and in 319-F, Figures 6(d) through (f), the eutectic Si particles have a finer distribution on the AS than RS. In 319-F, the Cu phases were also refined and distributed within the nugget zone. Microstructures (both grain and Si sizes) were more refined on the AS than RS due to the higher mechanical stirring rate on the AS. In the A390 alloys, Figures 6(g) through (i), both primary and eutectic Si, as well as Cu phases were refined. However, the originally coarser primary Si particles remain proportionally coarser than the eutectic Si particles after the FSW. A sharper thermo-mechanically affected zone (TMAZ) transition area was observed on the AS for all materials, as a result of different mechanical stirring rates and material flow patterns on the two sides of the nugget.

Several differences were observed in the FSW of the pre-weld heat-treated materials. Larger grain size was observed for higher rotation speeds in 6061-T651 due to a faster recrystallization rate, Figures 7(a) through (c). Similar grain sizes were observed in A356-T6 and 319-T6 at different rotation and traverse speeds, as nugget grain growth is limited by the lower heat input compared to 6061-T651 (lower rotation speeds) and the presence of Si particles. Nugget grain size after FSW is reduced by the pre-weld heat treatment in the cast alloys A356-T6, Figures 7(d) through (f), and 319-T6, Figures 7(g) through (i). As hardness of the base materials increased after the T6 heat treatment, the welding force and deformation rate during FSW have slightly increased. The higher deformation rate led to more refined grain structures in the nugget of pre-weld heat-treated A356-T6 and 319-T6 welds compared to their as-cast equivalents. Due to the presence and distribution of Si particles, nugget grain growth is prohibited in these alloys compared to 6061-T651, which has limited secondary phases. The Si particles in the weld nuggets of A356-T6 and 319-T6 were also slightly more refined in the pre-weld heat-treated conditions compared to the as-cast conditions due to the thermal modification during heat treatment that created smaller starting eutectic Si particles, as seen in Table II. Table VI summarizes the FSW microstructural results for all studied alloys without and with pre-weld heat treatment.

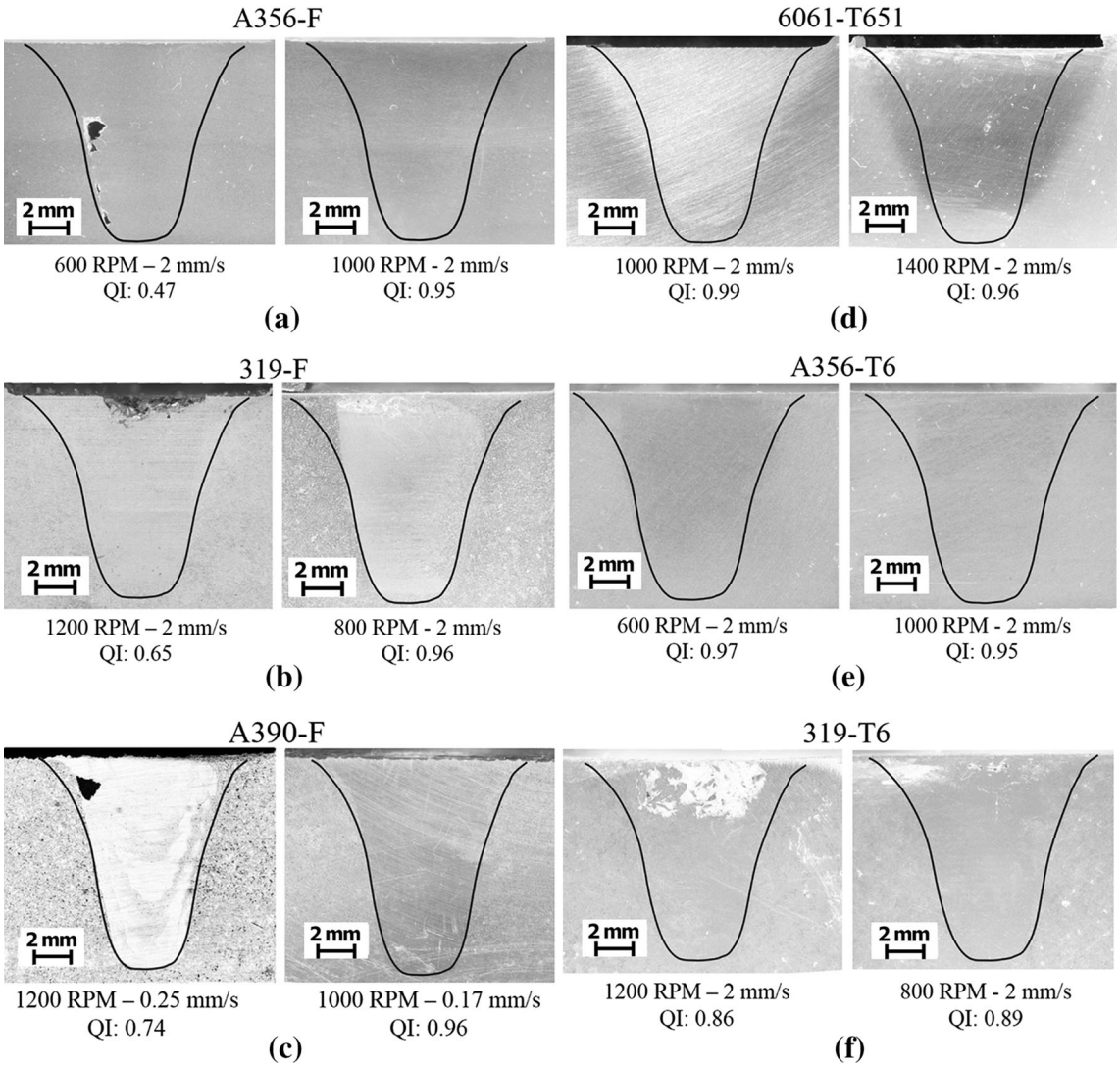


Fig. 5—FSW cross sections and weld quality indices for the as-cast alloys: (*a*) A356-F, (*b*) 319-F, (*c*) A390-F, and T6 heat-treated alloys: (*d*) 6061-T651, (*e*) A356-T6, and (*f*) 319-T6.

Band spacing (*i.e.*, width of the ''onion rings'' layers) was also measured for all tested materials and conditions. Each band represents a layer of material that has been ''extruded'' around the pin per rotation. The smaller the band spacing the faster the material flow during processing. Band spacing decreases with rotation speed and increases with traverse speed according to Eq. [9].

$$\text{Band spacing} = \frac{\text{Traverse speed} \times 1000}{\text{Rotation speed}/60}. \qquad [9]$$

In Eq. [9], the rotation speed is in (*RPM*) and the traverse speed in (mm/s). The band spacing gradually decreases from the center to the edge of the nugget on both AS and RS. Measurements were taken at the middle point between nugget center and edge on the AS (6 mm below the weld surface), which were considered as average values for the weld. Figures 8 and 9 show band spacing values for various alloys without and with pre-weld heat treatment under different combinations of rotation/traverse speeds, which are consistent with the theoretical values calculated from Eq. [9].

The band spacing of A356-F increases from ~152 μm, Figure 8(a), to ~244 μm, Figure 8(b), when the traverse speed increases from 2 to 3 mm/s. The band spacing decreases from ~244 μm, Figure 8(b), to ~164 μm, Figure 8(c), when the rotation speed increases from 800 to 1000 RPM. The band spacing values of 319-F, Figures 8(d) through (f), show similar trends to A356-F for changes in traverse and rotation speeds. The band spacing of 6061-T651 behaves similarly too, but is generally smaller due to the higher rotation speeds for this material, Figures 9(a) through (c). The measurements show that the largest band spacing is ~129 μm, under processing parameters of 1400 RPM—3 mm/s, Figure 9(b). In general, band spacing is smaller when FSW is done at lower traverse speeds and higher rotation speeds, as seen in Figures 8 and 9.

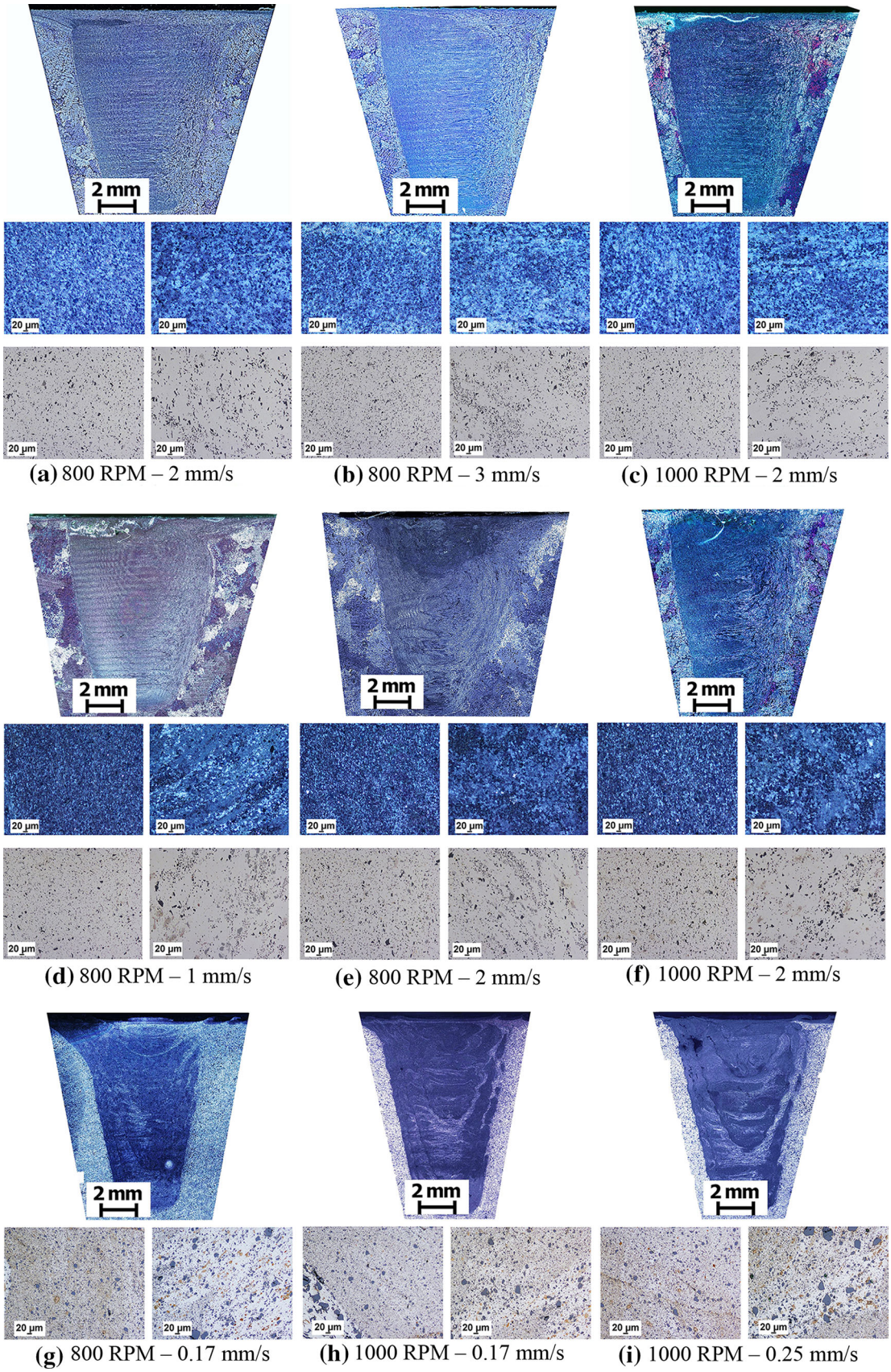


Fig. 6—Grain structures and characteristic phases in FSW nugget for: (*a* through *c*) A356-F, (*d* through *f*) 319-F, and (*g* through *i*) A390-F under various processing conditions.

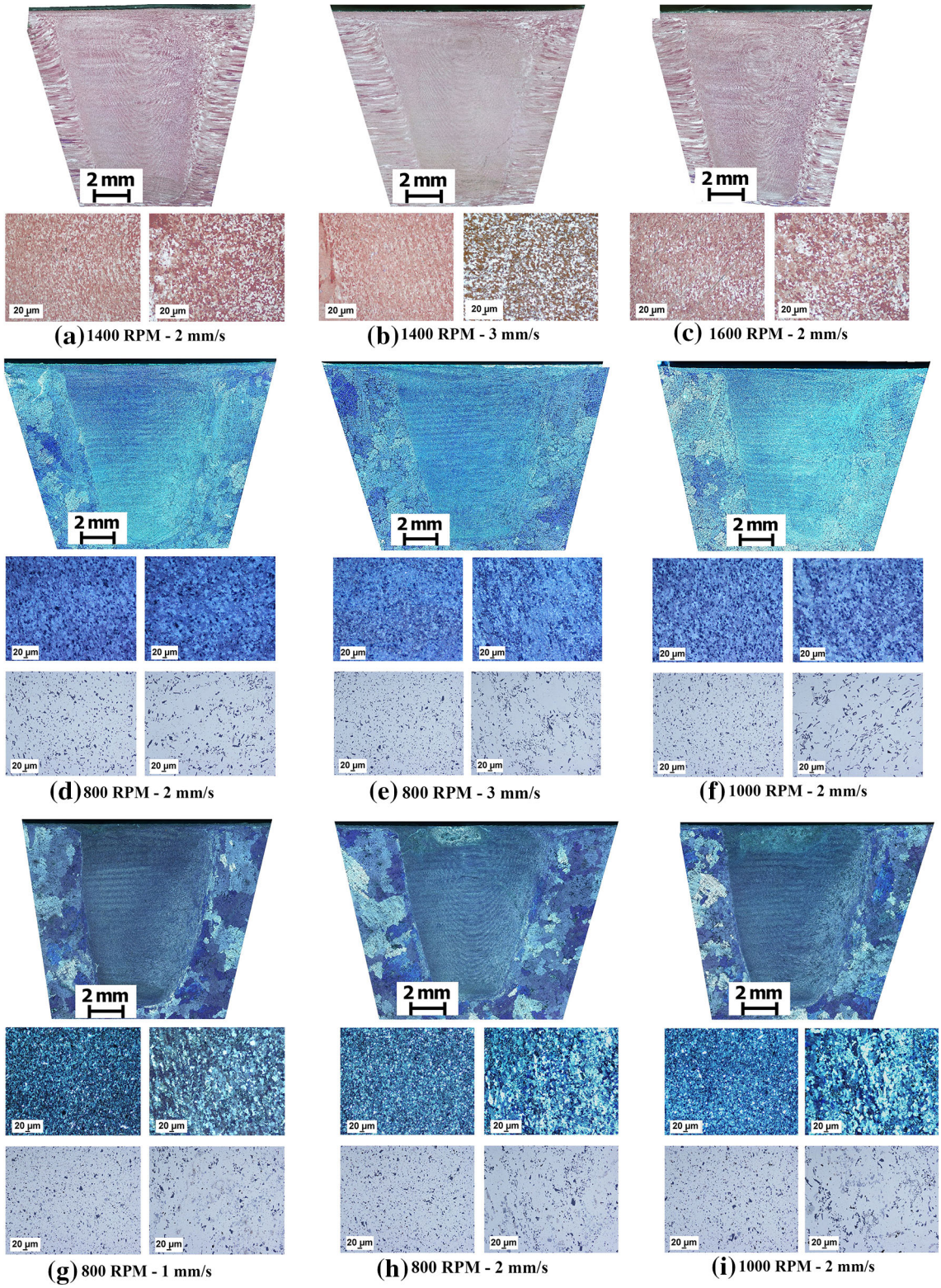


Fig. 7—Grain structures and characteristic phases in FSW nugget for: (*a* through *c*) 6061-T651, (*d* through *f*) A356-T6, and (*g* through *i*) 319-T6 under various processing conditions.

The pre-weld heat treatment on A356-T6 and 319-T6 does not change significantly the structure of the band spacing compared to the FSW in as-cast conditions, Figure 8(a) *vs* Figure 9(d), Figure 8(b) *vs* Figure 9(e), and Figure 8(c) *vs* Figure 9(f) for A356; and Figure 8(d) *vs* Figure 9(g), Figure 8(e) *vs* Figure 9(h), and Figure 8(f) *vs* Figure 9(i) for 319.

Table VI. Grain and Si Particle Sizes in the Nugget After FSW Without and With Pre-weld Heat Treatment

A356-F			319-F			A390-F		
RS (RPM)—TS (mm/s)	GS* (μm)	Eutectic Si (μm)	RS (RPM)—TS (mm/s)	GS (μm)	Eutectic Si (μm)	RS (RPM)—TS (mm/s)	GS (μm)	Primary Si (μm)
800—2	14.5	13	800—1	15	12	800—0.17	13	38
800—3	15	13	800—2	14	12	1000—0.17	14	38
1000—2	14	13	1000—2	12	11	1000—0.25	13	37

6061-T651		A356-T6			319-T6		
RS (RPM)—TS (mm/s)	GS (μm)	RS (RPM)—TS (mm/s)	GS (μm)	Eutectic Si (μm)	RS (RPM)—TS (mm/s)	GS (μm)	Eutectic Si (μm)
1400—2	14	800—2	13.5	13	800—1	7	12
1400—3	15	800—3	12	12	800—2	7	10
1600—2	17	1000—2	12.5	13	1000—2	9	10

* GS: nugget grain size.

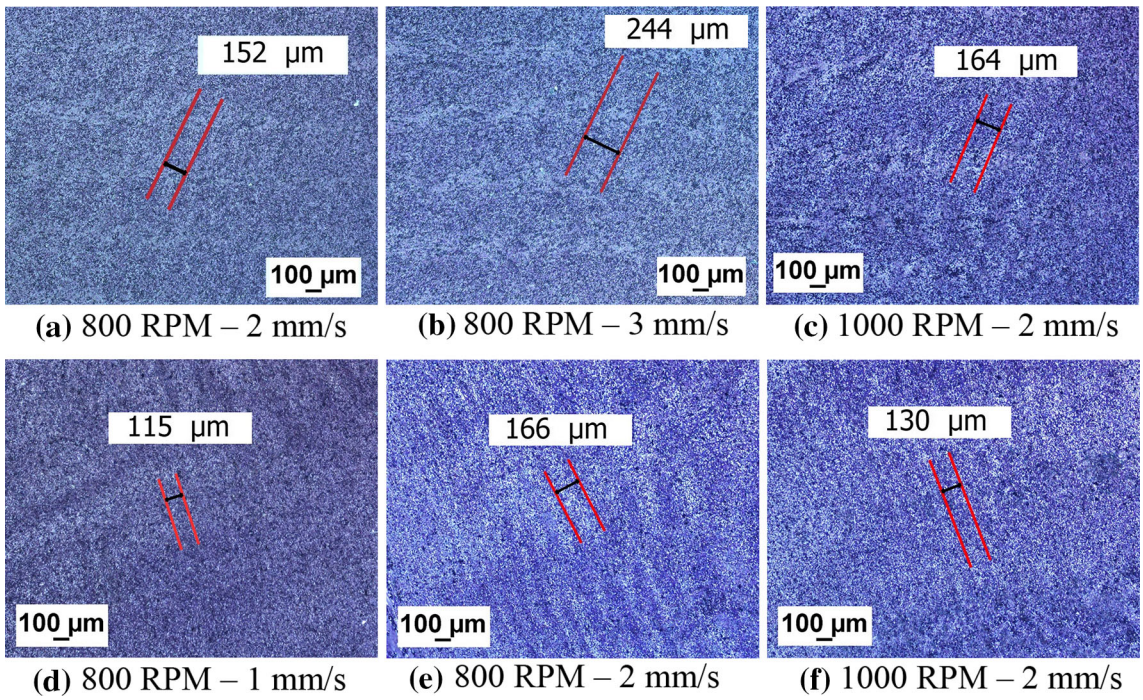


Fig. 8—Band spacing measurements in as-cast conditions for: (*a* through *c*) A356-F and (*d* through *f*) 319-F.

D. *Macro-/Micro-hardness*

Macro-/Micro-hardness tests were conducted on the cross-sectional area of the welds, at consistent locations. Vickers hardness (HV) micro-hardness measurements were performed 6 mm below the top surface using a 100 g load for 10 s, from the base material to the nugget on both sides of the processed zone. Brinell macro-hardness tests, using a 100 kg load *via* a spherical steel indenter of 1.6 mm in diameter, were conducted too, and the measurements converted to the HV scale. The same positions were used for both micro- and macro-hardness tests on two samples from the same weld.

Hardness profiles for all studied alloys in both as-cast and pre-weld heat-treated conditions were generated and are shown in Figures 10 and 11, respectively. For the as-cast alloys, A356-F, Figures 10(a) through (c), 319-F, Figures 10(d) through (f), and A390-F, Figures 10(g) through (i), an "A-shaped" hardness profile was observed across the FSW zone (the nugget hardness increases compared to that of the base materials). The peak values are reached in the center of the nuggets. The nugget hardness was increased by grain refinement, which corresponds to the Hall–Petch relationship in Eq. [10], and more importantly, by the refinement of different secondary phases. In A356, the hardness increase was related to both eutectic Si and Fe-containing phases. Due to the high content of Cu in both 319 and A390 (and the high content of Si in A390), the hardness was more

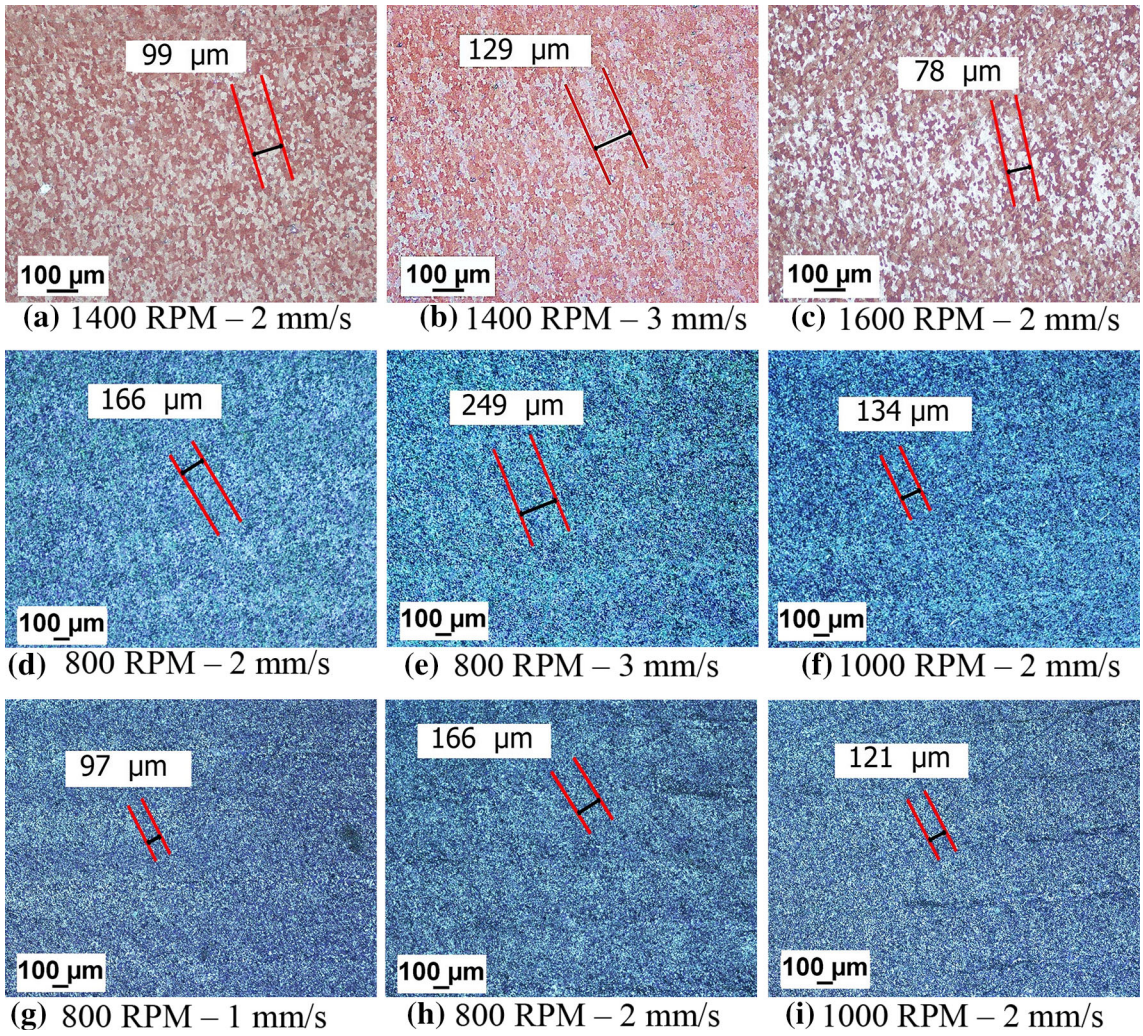


Fig. 9—Band spacing measurements for pre-weld heat-treated samples: (*a* through *c*) 6061-T651, (*d* through *f*) A356-T6, and (*g* through *i*) 319-T6.

affected (higher) in these alloys by the redistribution of the secondary phases *via* mechanical stirring. The reduced size of the secondary phases and their closer distribution will increase the nugget hardness, while the α-Al matrix will gain ductility due to the elimination of porosity. Despite the fact that the Si content is similar in A356 and 319 alloys, the presence of Cu phases in 319 has a more pronounced effect on the increase in the nugget hardness of this alloy, as seen in Figures 10(d) through (f) compared to Figures 10(a) through (c). Nugget hardness is also affected by the processing parameters; it increases with higher rotation and traverse speeds, as a result of a more refined distribution of secondary phases with higher mechanical stirring rates. A slight decrease was observed in the HAZ after FSW in all cases.

$$\sigma_y = \sigma_0 + \frac{k}{\sqrt{d}}. \qquad [10]$$

In the T6 pre-weld heat-treated conditions, the nugget hardness decreased compared to the base material for 6061, Figures 11(a) through (c) and A356 alloys, Figures 11(d) through (f). In these materials, a "W-shaped" hardness profile was observed across the FSW zone, with hardness values in nugget zone, TMAZ, and HAZ lower than in the base materials. In both alloys, the two regions of minimum hardness are located in the HAZs. Lower hardness in both nugget and HAZ, and wider HAZ were observed for high rotation speeds compared to low rotation speeds. This is explained by higher recrystallization rates, and therefore larger grains associated with more heat input in the high rotation speed cases, as shown in prior FSW thermal modeling work by Pan and Lados.[36] The hardness decreases in HAZ are directly related to the amount of heat input during FSW. As for all FSW conditions, the nugget zone temperature is much higher than the precipitate dissolving temperature, the precipitates will have similar size and effect on the resulting nugget hardness. Thus, the hardness differences observed between various processing conditions are more related to grain size refinement and residual stresses, which are influenced significantly by the dynamic recrystallization rate during FSW. In the pre-weld heat-treated 319-T6, Figures 11(g) through (i), an "A-shaped" hardness profile across the

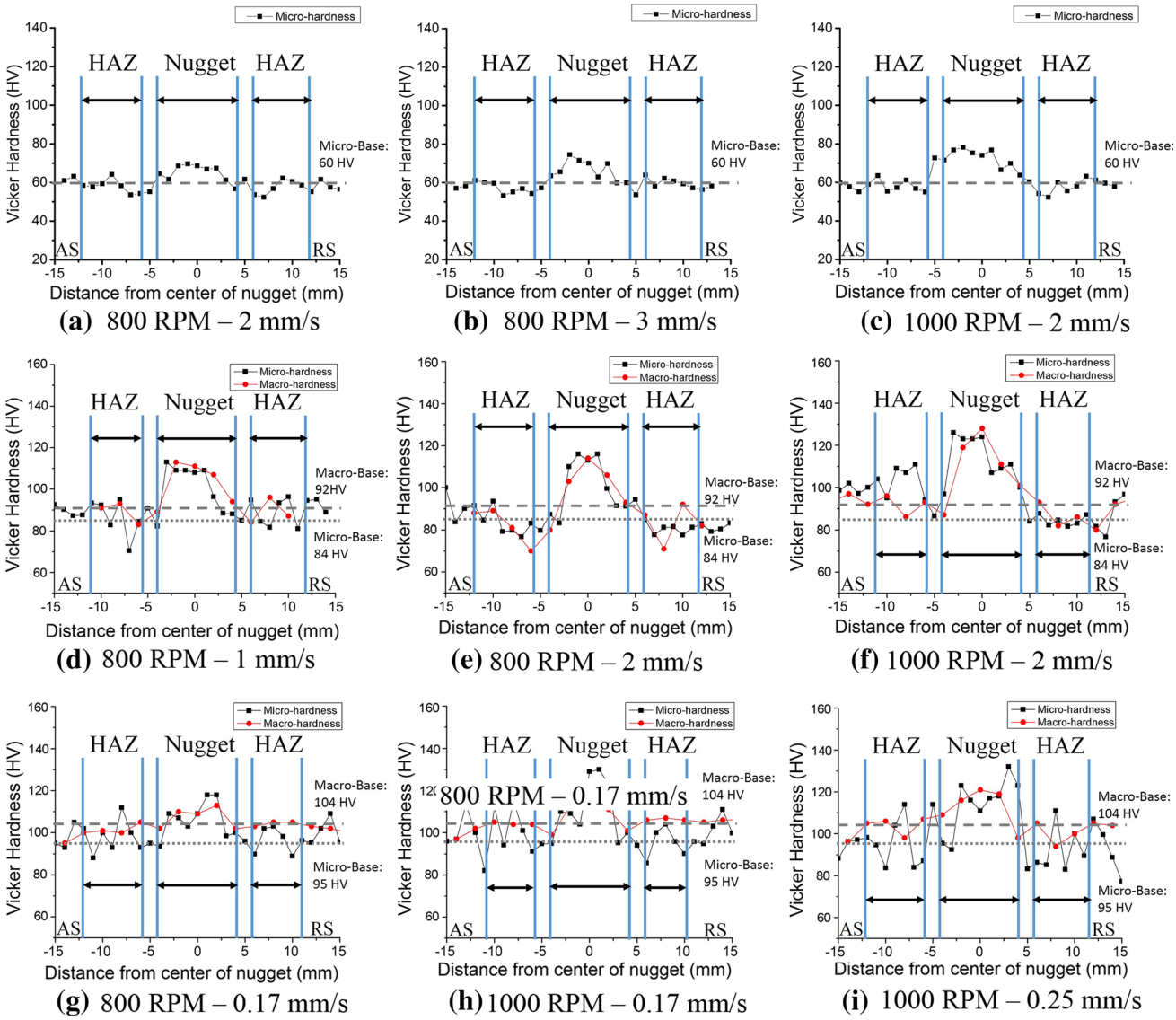


Fig. 10—Hardness profiles across FSW zones for the cast alloys processed in as-cast conditions: (*a* through *c*) A356-F, (*d* through *f*) 319-F, and (*g* through *i*) A390-F.

FSW zone was observed, where the nugget hardness is higher than that of the base material.

The volume fractions of the secondary phases (Fe phase in A356-T6 base material and Cu phase in 319-T6 base material) were measured at ten random locations, Table VII, and the average values were found 0.8 pct for the Fe phase and 6.1 pct for the Cu phase. The secondary phase contents of A356-T6 and 319-T6 were further analyzed with Pandat™ at room temperature, Table VIII, and the results show consistency with the measurements (1.2 pct for the Fe phase in A356-T6 and 7.5 pct for the Cu phase in 319-T6). Due to the similar eutectic Si content in both alloys (6.4 pct in A356 and 5.8 pct in 319), the redistribution of refined Cu phase (6.1 pct) largely controls the hardness in 319, which explains why the nugget hardness of this alloy is higher than that of the base material, and different from that of A356-T6, which has a smaller amount of Fe phase (0.8 pct) and other reinforcing secondary phases, all totaling ~2.5 pct.

The precipitation hardening effects were further analyzed in A356-T6 and 319-T6 (A356-T6 and 6061-T651 share the same precipitate system and consequent hardness changes). The calculations were based on Zener's theory of precipitate growth.[37] Several assumptions were made for the model:

- A spherical precipitate of radius R was assumed to grow in the α-Al matrix;
- Initially, an isotropic growth was considered for the coherent precipitates;
- Uniform distribution of the chemical composition is achieved inside each precipitate;
- α-Al matrix is considered to be of infinite size compared to the precipitate size, and no interaction occurs between precipitates;

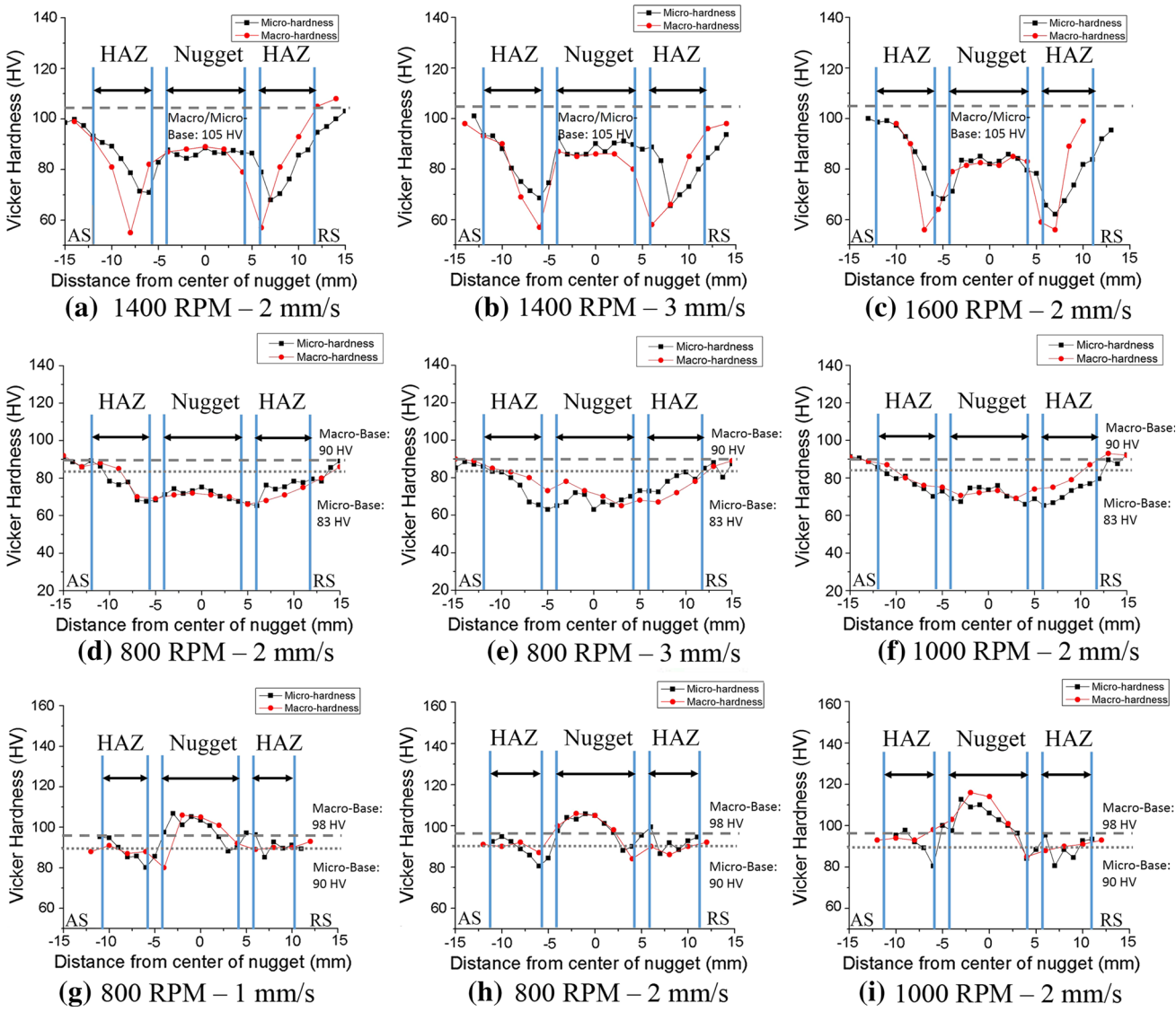


Fig. 11—Hardness profiles across FSW zones of the pre-weld heat-treated alloys: (*a* through *c*) 6061-T651, (*d* through *f*) A356-T6, and (*g* through *i*) 319-T6.

Table VII. Typical Secondary Phase Contents in A356-T6 and 319-T6 Alloys

	Location (pct)										
Phase	1	2	3	4	5	6	7	8	9	10	Average
Fe phase (A356-T6)	1.2	0.2	0.9	0.4	1.4	0.5	2.9	0.1	0.2	0.5	0.8
Cu phase (319-T6)	5.7	6.5	2.9	5.4	7.4	6.9	8.7	5.6	5.2	6.5	6.1

Table VIII. Secondary Phase Contents Simulated with Pandat™ for Alloys A356-T6 and 319-T6

A356 (T6)	α-Al (pct)	Eutectic Si (pct)	Al_2Cu (pct)	AlFeCuMgSi (pct)	Al_5FeSi (pct)
Fraction	90.9	6.4	0.3	1.2	1.2

319 (T6)	α-Al (pct)	Eutectic Si (pct)	Al_2Cu (pct)	Mg_2Zn_{11} (pct)	Al_5FeSi (pct)	$Al_{15}(FeMn)_3Si_2$ (pct)
Fraction	81.8	5.8	7.5	0.7	1.4	2.8

- The partial molar volumes of precipitate phase and matrix are the same, which ensures no distortion at the boundary.

In A356-T6, β' and β'' phases of Mg_2Si are assumed, and in 319-T6, Cu exists in forms of θ' and θ'' phases. All the metastable phases in both alloys are primarily controlled by the thermal cycle rather than the mechanical stirring. Both diffusion coefficients of Mg and Cu in the Al matrix are assumed to be only a function of temperature. The equilibrium concentration at the interface between the precipitate and matrix is achieved at all times. The concentration of Mg (or Cu) in matrix is C_0, and the equilibrium concentration close to the interface between the precipitate and matrix is C_e, both of which are assumed constant in the calculations. The concentration of Mg (or Cu) within the precipitate particles is C_p. The characteristic diffusion layer ahead of the precipitate interface is assumed of a thickness corresponding to the size of the precipitate. The precipitate growth model is given in Eq. [11].[37]

$$\frac{dR}{dt} = \frac{D}{C_p - C_e} \cdot \left.\frac{\partial C}{\partial r}\right|_R . \qquad [11]$$

The equation was further simplified into Eq. [12], with an estimated diffusion gradient at the interface, by making $\left.\frac{\partial C}{\partial r}\right|_R \cong \frac{C_0 - C_p}{L}$. The precipitate radius R was then calculated by integration with time.

$$\frac{dR}{dt} = \frac{D}{C_p - C_e} \cdot \frac{C_0 - C_p}{L} . \qquad [12]$$

The intrinsic diffusion coefficient of Cu in Al and the activation energy were based on Du *et al.*'s research,[38] $D_0 = 4.44 \times 10^{-5} m^2/s$, $Q = 133.9\,KJ/mole$. The diffusion coefficient of Mg in Al at 473 K (200 °C) is $\sim 0.5 \times 10^{-13} m^2/s$.[39] The calculated diffusion coefficient of Cu in Al at 473 K (200 °C) is $\sim 7.5 \times 10^{-12} m^2/s$. So β'' should have a smaller growth rate than θ''. The elemental contents in both precipitate and matrix were determined from stoichiometric calculations and EDS chemical composition analysis, respectively. The calculated concentrations are

$$C_{p-Mg} = 63.16\,wt\,pct\,Mg\,(in\,Mg_2Si\,of\,A356)$$

$$C_{0-Mg} = 0.3\,wt\,pct\,Mg\,(in\,\alpha - Al\,matrix\,of\,A356)$$

and

$$C_{p-Cu} = 54.23\,wt\,pct\,Cu\,(in\,Al_2Cu\,of\,319)$$

$$C_{0-Cu} = 3.63\,wt\,pct\,Cu\,(in\,\alpha - Al\,matrix\,of\,319)$$

The equilibrium concentration, C_e, was assumed 1 wt pct. The temperature used in the calculations was determined from our prior work on heat transfer modeling.[36] The radius change was numerically integrated, which resulted in a precipitate size change of approximately 0.1 nm. Compared to these coherent precipitates, incoherent and semi-coherent interfaces will have much lower activation energy and faster growth rates. This will lead to precipitate coarsening of several nanometers, which is sufficient to cause a loss of strength in high energy input welding conditions. Considering the thermal history during FSW, with a peak temperature of approximately 473 K (200 °C) at 800 RPM and 2 mm/s, the coarsening of β'' and θ'' is limited. In some higher temperature cases,[27] the peak temperature could reach ~673 K (400 °C), in which case the coarsening will be much faster, further decreasing the hardness in HAZ.

E. *Tensile Properties Within and Across the Weld Nugget*

Tensile tests were performed according to the ASTM-E8 standard. Rectangular tensile specimens were extracted from both longitudinal (within) and transverse (across) directions of the processing zones, as shown in Figure 12. The dimensions of the tensile bars were 25.4 mm (gage length), 5 mm (gage width), 6.5 mm (fillet radius), 13 mm (grip length), 13 mm (grip width), and 5 mm (thickness). For the longitudinal specimens, the whole gage area was within the nugget. For the transverse specimens, the center of the nugget was located at the middle of the gage length.

All samples extracted within the nugget underwent a large increase in ductility after FSW compared to the base materials, Figure 13. In the as-cast alloys, the ductility increased 2 to 3 times compared to the base materials' ductility, Figures 13(a) through (c). For the pre-weld heat-treated cases, the ductility doubled in 6061-T651, Figure 13(d), and increased more than ten times in A356-T6 and 319-T6, Figures 13(e) and (f). The increase in ductility for all cases is due to the significant grain and phase refinement in the nugget zone. Given that A356-T6 and 319-T6 base materials were pre-weld heat-treated to peak aging conditions, they both start with higher tensile strength and lower ductility compared to the as-fabricated conditions. Also, in the base materials, porosity and large eutectic Si and other secondary phases cause significant stress concentration under tensile loading. After FSW, grains and all secondary phases were significantly refined in the nugget zone, which leads to low stress concentrations and more uniform stress distributions. All these concomitant phenomena contribute to significant ductility increases.

Tensile strength within nugget zone behaves differently depending on the type of alloys and pre-weld heat treatment conditions. In most as-cast alloys, both yield strength (YS) and ultimate tensile strength (UTS) increased due to the increase in grain refinement, Figures 13(a) through (c). The pre-weld heat treatment increases the tensile strength of the base materials. After FSW, 6061-T651 and A356-T6, experience a decrease in the YS and UTS, with a significant recovery of ductility as mentioned before, Figures 13(d) and (e). In 319-T6, the ductility could not be measured and compared to that of FSW in as-cast conditions, Figure 13(f). However, the

trends in both ductility and strength after FSW for 319-T6 are expected to be similar to those in A356-T6.

Rotation and traverse speeds also contribute to changes in tensile properties. Higher rotation speeds provide more thermal energy to the dynamic recrystallization and a slightly larger nugget grain size, which generally results in lower UTS values, Figure 13. Higher traverse speeds increase the UTS by lowering the thermal energy input and decreasing the grain size. Higher traverse speeds also introduce faster plastic deformation rates during FSW, which increases the dislocation and subgrain structure density. This mechanism constrains lattice sliding, which increases the UTS of the welds.

Secondary phase refinement also contributes to the increase in the tensile strength of cast alloys. In A356, the eutectic Si particles and Fe phases were significantly reduced in size. These particles were uniformly distributed within the nugget by the mechanical stirring, lowering the stress concentration. In 319, the refined eutectic Si particles and Cu phases both improved the increase in YS and UTS. In A390, due to the coarser size of the primary Si particles, the nugget zone show less increase in strength compared to the other cast alloys, despite the fragmentation of the primary Si phase.

Tensile properties across the nugget are primarily controlled by the strength of HAZ and TMAZ. Ductility of FSW as-cast alloys (A356-F and 319-F, Figures 14(a) and (b)) is similar to that of the base materials (A390-F failed with very low ductility). For the pre-weld heat-treated FSW alloys, ductility decreased in 6061-T651, Figure 14(c), and remained unchanged for A356-T6 and 319-T6, Figures 14(d) and (e). The decrease of strength in 6061-T651 is mainly due to the weaker HAZ. For the cast alloys (both as-cast and T6 conditions), the tensile properties in the across-nugget direction are mainly dictated by the base materials' properties and casting defects.

Fracture surfaces of the tensile specimens across the weld were analyzed, and the results for 6061-T651 and A356-F are shown in Figure 15. Two failure mechanisms were observed. In the 6061-T651, failures occurred mostly on the AS in the HAZ, which has the

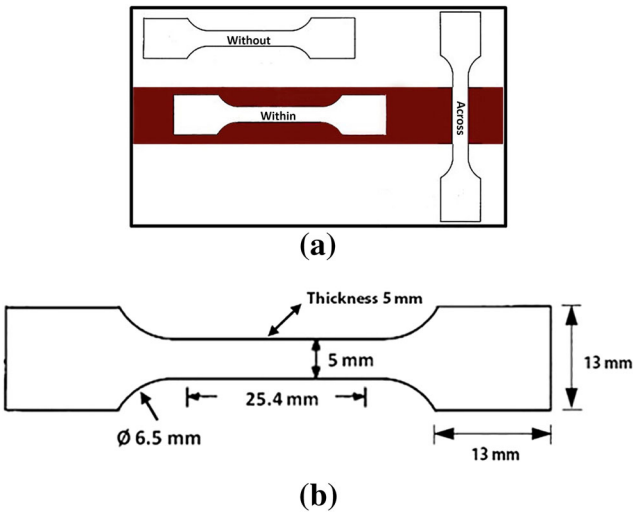


Fig. 12—Layout (*a*) and dimensions (*b*) of the tensile specimens.

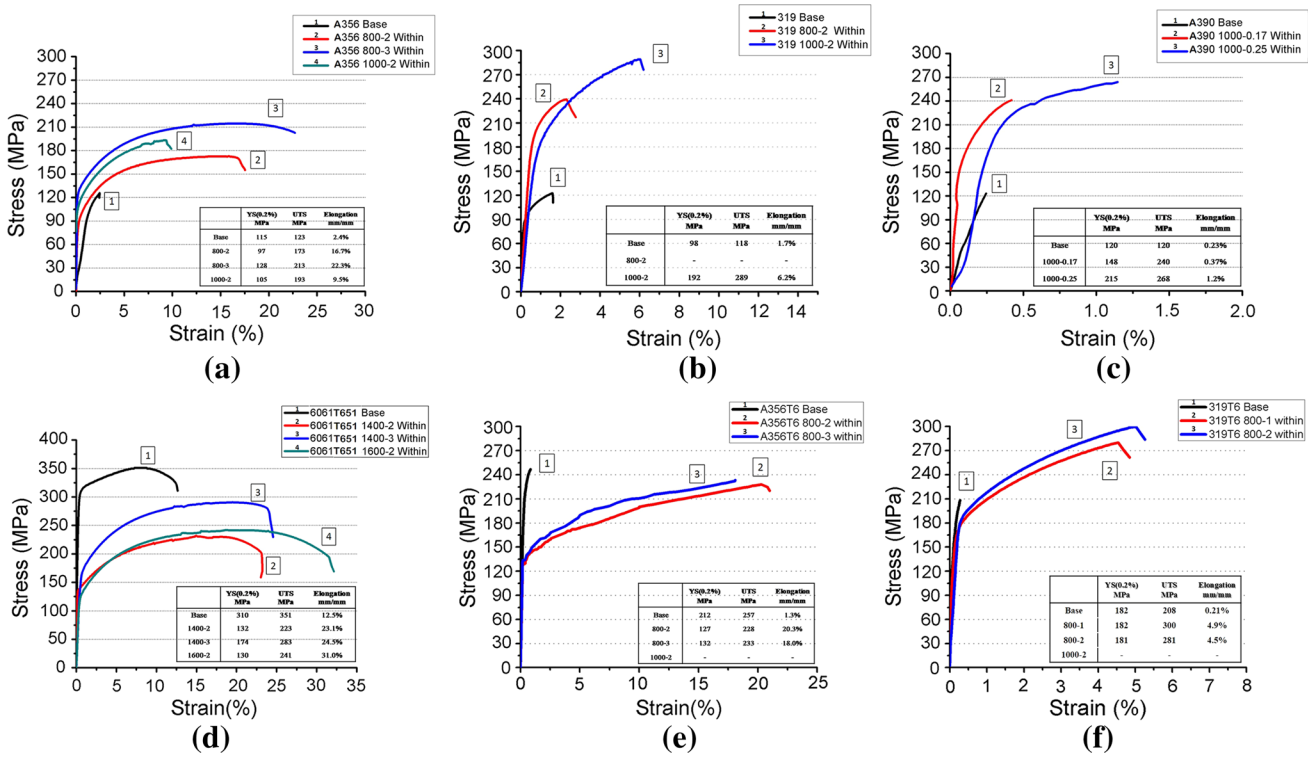


Fig. 13—Tensile properties within the welds for (*a* through *c*) as-cast and (*d* through *f*) pre-weld heat-treated alloys.

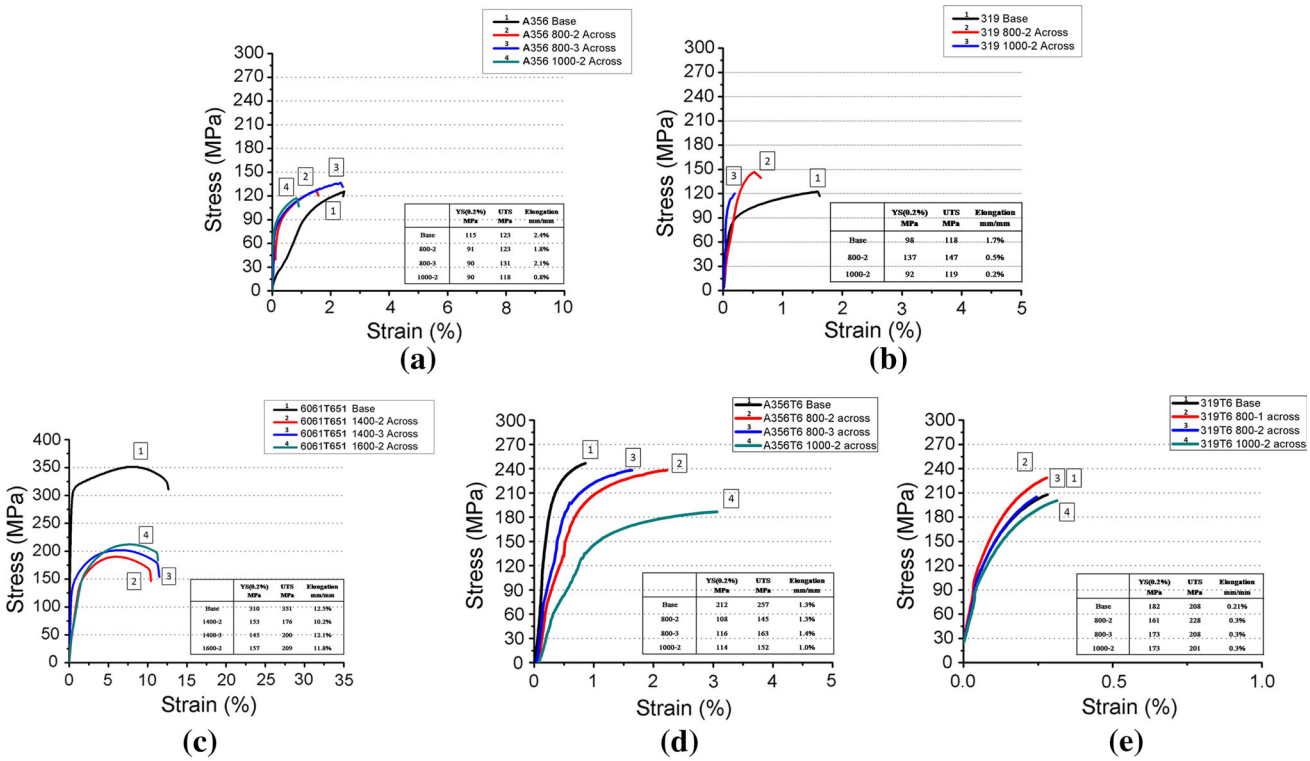


Fig. 14—Tensile properties across the welds for (*a*, *b*) as-cast and (*c* through *e*) pre-weld heat-treated alloys.

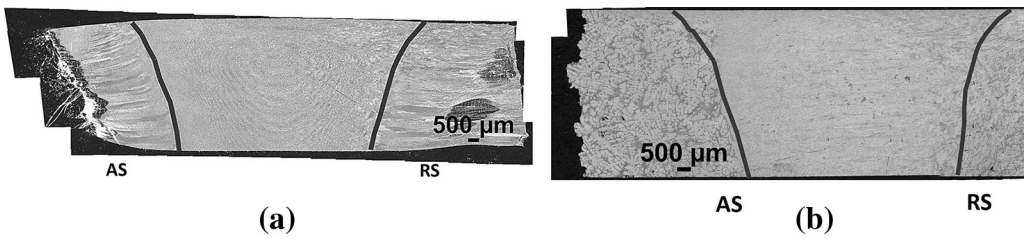


Fig. 15—Fracture locations in tensile specimens for (*a*) 6061-T651 and (*b*) A356-F.

lowest hardness across the entire processed zone. On this side, the boundary between HAZ and TMAZ has a sharper transition area too, which causes less uniform stress distribution. Thus, controlling the area and properties of the HAZ plays an important role in increasing the strength of the welds. The angle between the fracture surfaces and normal direction is close to 45 deg where the principle shear stress direction is aligned, Figure 15(a). Severe necking and elongation were observed in the HAZ on both AS and RS, but no deformation occurred in the nugget. In the cast alloys, failure often initiated in the base material at casting defects, with the fracture surface being almost perpendicular to the loading direction, Figure 15(b). This limits the overall mechanical properties of the FSW cast materials, and tensile results in the across-nugget direction indicate that HAZ is not controlling the weld failure as damage occurs at typical defect locations in the base material. In the wrought alloys, failure is related to the HAZ, and weld properties can be improved by controlling the size/properties of the HAZ using optimized processing conditions.

IV. CONCLUSIONS

A. *Weld Quality Model*

A weld quality model was successfully developed based on the stress concentration and location of welding defects. Defects morphology with aspect ratio close to 1 (round) has lower stress concentration, while defects with lower aspect ratio (elongated) have higher stress concentration and are more detrimental when the weld is subjected to loads in perpendicular direction. The weld quality index varies from 0 to 1, indicating a corresponding increase in weld quality and expected performance. The welds are classified into four levels based on the quality index (poor, fair, good, and

perfect), and acceptable welds (good and perfect) are considered to have index values above 0.8 and 0.9, respectively.

B. *Process Optimization*

For the as-cast A356-F and 319-F hypoeutectic alloys, the appropriate processing ranges are 800 to 1000 RPM for the rotation speed and 2 to 3 mm/s for the traverse speed. For A390-F hypereutectic alloy, the appropriate rotation speeds are similar to the other cast alloys, but with traverse speeds less than 0.25 mm/s due to the highest Si content. Weld quality generally decreases with increases in the traverse speed in the cast alloys.

Appropriate processing parameters for 6061-T651 combine a wider a rotation speed range, from 800 to 1600 RPM, with traverse speeds from 1 to 3 mm/s. The pre-weld heat treatment also significantly increased the processing range for both A356-T6 and 319-T6 alloys. Specific results and ranges may vary for tests conducted with different fixtures and tool geometries.

C. *Microstructures of the Welds*

All alloys show significant grain refinement in the nugget (DXZ). Finer grain size and Si particles (both eutectic and primary Si) were observed on the AS due to higher stirring rate. Band spacing in DXZ (''onion rings'') increases with traverse speeds and decreases with rotation speeds.

D. *Hardness*

In all as-cast alloys, an ''A-shaped'' hardness profile was observed, with higher nugget hardness. The increase in nugget hardness is due to the refinement of eutectic Si and Fe-rich phases in A356-F; eutectic Si and Cu phase in 319-F; and both eutectic and primary Si and Cu phase in A390-F. Alloys 6061-T651 and A356-T6 display ''W-shaped'' profiles with the lowest hardness in the HAZ due to loss of precipitate hardening. In the pre-weld heat-treated A356-T6, the nugget hardness shows an opposite trend (lower nugget hardness) compared to A356-F, while 319-T6 and 319-F show the same trend (higher nugget hardness) due to the dominant effects of hard Cu phases in this alloy.

E. *Tensile Properties*

Tensile properties within the nugget (along the welding direction) are improved by grain and secondary phase refinement in all alloys. In across-nugget direction, tensile properties are determined mainly by the strength of the HAZ in 6061-T651, while in the cast alloys are dictated by the base material microstructure and properties. For 6061-T651, the fracture occurs in the HAZ due to the loss of precipitation strengthening, and for cast Al alloys at pre-existing defects in the base material.

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